

TELECOM CHURN ANALYTICS

Power BI + Python ML

5000 customers | 133k rows | 7 tables

Churn 42.9% | Acc 71.3% | AUC 0.7789

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1. What is this project? In 30 seconds

Phone companies lose money when customers leave. This is called churn. We built fake but real-like data for 5000 Egypt telecom customers, then we answer 3 questions: Who leaves? Why? Who will leave next?

- Data: 7 tables. Think like Excel sheets linked by customer_id.
- Dashboard: Power BI shows churn with charts and filters.
- Future: a smart model gives each customer a risk score 0-100%.

2. The 7 tables - super simple

customers: who they are (age, city, contract). subscriptions: what plan they buy (Basic/Plus/Ultra/Business, Fiber/DSL/5G, price, months). usage_monthly: what they use each month (calls, GB, downtime). billing: do they pay late? support_tickets: do they complain? network_quality: is signal bad in their region? churn_labels: did they leave + why.

Link: customer_id is the key. One customer has 12 usage rows + 12 billing rows + 0-5 tickets. This is called star schema. It makes Power BI fast.

3. Power BI dashboard - 5 pages

Page 1 Overview: big numbers (KPIs) + churn trend. Page 2 Customer: churn by plan/contract/payment. Page 3 Usage+Network: outages vs churn, map by region. Page 4 Billing+Support: late pay + satisfaction. Page 5 Prediction: high-risk list + money at risk.

DAX is just Excel formulas for Power BI. Example: $\text{Churn Rate} = \text{Churned} / \text{Total} * 100$. All formulas are in powerbi/DAX_measures.dax. Copy-paste them.

Steps: Load 7 CSVs, fix types with PowerQuery_M.m, make links, paste DAX, drag charts. 10 minutes. See powerbi/data_model_and_dashboard.md.

4. The ML model - like a smart friend

We teach the computer with old data: these customers left, these stayed. The computer finds patterns. Then for a new customer it says: 82% risk, take care!

Model: RandomForest, 200 trees. Each tree votes leave/stay. Simple and strong. No deep learning needed.

- Score: accuracy 71.3% (71 of 100 correct), AUC 0.7789 (0.5 = random, 1.0 = perfect).
- Top drivers: bill amount, monthly charge, usage, tenure, downtime, late days, tickets, satisfaction.
- Use it: `py ml/predict.py`. You get `churn_probability` + Low/Medium/High.

5. Results - what we learned

Churn rate 42.9% (2143 of 5000). Month-to-month churns 2x more than 2-year. Fiber has more tickets. Electronic check + paperless = more churn. Low satisfaction (1-2 stars) churns a lot. New customers (tenure under 6 months) are most risky.

Money tip: call High-risk customers first, give discount or fix network, move them to 1-year contract. This saves lifetime money (CLV).

6. How to run + files

Windows: double-click `run.bat`. Or: `pip install -r requirements.txt`, `py make_dataset.py`, `py build_analysis_ml.py`, `py ml/predict.py`. Preview: `dashboard_preview/index.html`. Charts: `analysis/charts` (13 images).

Folders: `data` = CSVs, `powerbi` = DAX+M+plan, `ml` = `model.pkl`+metrics+predict, `analysis` = charts, `docs` = this PDF + references.

7. References (where to learn more)

- Microsoft Power BI dashboards - learn.microsoft.com/power-bi/create-reports/
- DAX overview - learn.microsoft.com/dax/dax-overview
- Power Query M - learn.microsoft.com/powerquery-m/
- Star schema - learn.microsoft.com/power-bi/guidance/star-schema
- pandas docs - pandas.pydata.org/docs/
- scikit-learn RandomForest - scikit-learn.org/stable/
- IBM Telco Churn classic dataset - inspiration for columns
- Google Scholar: customer churn prediction telecom review

I used first 4 for dashboard, next 2 for code, last 2 for churn idea. Data here is my own fake Egypt data, bigger: 7 tables, 133k rows.